

Warm-up 1

Write down the matrix that sends $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and that sends $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

Where does this matrix send the vector $\begin{pmatrix} 10 \\ 0 \end{pmatrix}$? What about the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$?

Warm-up 2

Compute the following real matrix-vector product:

$$\begin{pmatrix} 8 & -2 & 4 & 23 & 6 \\ 0 & 4 & 31 & 2 & 8 \\ -5 & 4 & -2 & 4 & 20 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} =$$

What are the domain and codomain of the linear transformation defined by the matrix?

During class 1

a) Compute in $\mathbb{Z}/5\mathbb{Z}$:

$$4 + 3 =$$

$$2 - 3 =$$

$$3 \times 4 =$$

b) Compute all of the scalar multiples of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in $(\mathbb{Z}/5\mathbb{Z})^2$.

During class 2

Which of the following are vector spaces? For those that aren't vector spaces, explain which vector space axiom fails.

a) the vectors in $\mathbb{R}^2$ with non-negative entries

b) the solutions to the equation $x + 2y + 3z = 6$ over the real numbers

c) the solutions to the system of equations

$$x + 3y + 4z = 0$$

$$x + 2y + 3z = 0$$

where the variables take values in $\mathbb{R}$

And what if instead they take values in $\mathbb{C}$? Or $\mathbb{Z}/5\mathbb{Z}$?